Code No: 117FZ **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year I Semester Examinations, December - 2019
OPERATIONS RESEARCH
(Common to ME, CSE, MCT, MIE, AGE)

Time: 3 Hours **Max. Marks: 75**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A. Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

PART- A

(25 Marks)

1.a) What is Operations research? [2]
b) What are the limitations of LP problem? [3]
c) Formulate the travelling – Salesman problem as an assignment problem. [2]
d) What is a degenerate transportation problem and how do you resolve it? [3]
e) What are the situations which make the replacement of items necessary? [2]
f) Write three assumptions of job sequencing. [3]
g) What are the characteristics of game theory? [2]
h) What are the functions of inventory management? [3]
i) What are major limitations of simulation? [2]
j) What are the applications of dynamic programming? [3]

PART-B

(50 Marks)

2. Use penalty (or Big-M) method to maximize $z = 3x_1 - x_2$
Subject to the constraints $2x_1 + x_2 \geq 2$; $x_1 + 3x_2 \leq 3$; $x_2 \leq 4$ $x_1, x_2 \geq 0$ [10]

OR

3. A person requires 10, 12, and 12 units of chemicals A, B and C respectively for his garden. The liquid product contains 5 units, 2 units and 1 unit of A, B and C respectively per jar. The dry product contains 1 unit, 2 units and 4 units of A, B and C per jar. If the liquid product sells for Rs.3 per jar and dry product sells for Rs.2 per jar, how many of each should be purchased to minimize the cost and meet the requirements. [10]

4. Solve the following transportation problem by Least Cost method and check the feasibility of the solution. The unit costs of shipment are given in the cells below [10]

To / From	D	E	F	Supply
A	6	4	1	50
B	3	8	7	40
C	4	4	2	60
Demand	20	95	35	150

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OR

5. A company has one surplus truck in each of the cities A, B, C, D, and E and one deficit truck in each of the cities 1, 2, 3, 4, 5 and 6. The distance between the cities in Kilometers is shown in the matrix below. Find the assignment of trucks from cities in surplus to cities in deficit so that the total distance covered by the vehicles is minimum. [10]

	1	2	3	4	5	6
A	12	10	15	22	18	8
B	10	18	25	15	16	12
C	11	10	3	8	5	9
D	6	14	10	13	13	12
E	8	12	11	7	13	10

6. Find the Sequence for the following eight jobs that will minimize the total elapsed time for the completion of all jobs. Each job is processed in the same order ABC. The following entries give the time in hours on the machines. Find the total elapsed time and the idle line on each machine. [10]

Jobs (→) Times on Machines (↓)	1	2	3	4	5	6	7	8
A	4	6	7	4	5	3	6	2
B	8	10	7	8	11	8	9	13
C	5	6	2	3	4	9	15	11

OR

7. Find the cost period of individual replacement of an installation of 300 lighting bulbs, given the following:
   a) Cost of replacing individual bulb is Rs. 3. The cost of individual replacement is Rs 1.25.
   b) Conditional probability of failure is given below [5+5]

Week number :	0	1	2	3	4
Conditional probability of failure	0	1/10	1/3	2/3	0

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8. Solve the following game by dominance method. [10]

		Player A					
		B1	B2	B3	B4	B5	B6
Player B	A1	4	2	0	2	1	1
	A2	4	3	1	3	2	2
	A3	4	3	7	-5	1	2
	A4	4	3	4	-1	2	2
	A5	4	3	3	-2	2	2

OR

9.a) Explain Different inventories .

b) Find the economic lot size that associates with total cost and the length of time between two orders, given that the set-up cost is Rs.100, daily holding cost per unit of inventory is 5 paise and daily demand is approximately 30units. [4+6]

10. Jobs arrival at a workstation in a manufacturing plant is in a Poisson fashion at an average rate of five per hour. The time to machine one job is an exponential distribution with a mean time of 10 minutes. What is the expected time a job has to wait at the workstation? What will be the average number of jobs waiting at the workstation at any time? What is the probability that there will be more than four jobs? [10]

OR

11.a) State the Bellman's principle of optimality in dynamic programming and give a mathematical formulation of a dynamic programming problem?

b) Explain the Process of simulation and give its applications. [6+4]

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